

ADVANCED GUIDE: COFFEE, BOTANY, AND COGNITIVE PERFORMANCE

1. Origins and Botany Coffee originated in the highlands of Ethiopia, where the *Coffea* shrub grows wild. Although there are more than 100 species, the global trade is mainly divided between two: Arabica (*Coffea arabica*) and Robusta (*Coffea canephora*). Arabica represents 60-70% of world production, grows at high altitudes, and is characterized by acidic and complex flavors. Robusta, on the other hand, is more resistant to pests, has twice the caffeine content, and contributes a more bitter and earthy flavor, ideal for adding body to espresso.

2. The Roasting Process and Chemistry During roasting, green coffee beans undergo physical transformations. The most important milestone is the "First Crack," a thermal expansion where the bean releases water vapor and doubles in size, similar to popcorn. Chemically, the most relevant component is caffeine (1,3,7-trimethylxanthine). This molecule works by blocking adenosine receptors in the brain; adenosine is responsible for signaling sleepiness, so by blocking it, coffee keeps us alert and focused.

3. Brewing Methods and Pressure Brewing coffee is a matter of physics. Espresso requires 9 bars of pressure to force hot water through finely ground coffee, creating the "crema" (an emulsion of oils and air). Other popular methods include:

Flat White: Originating in Oceania, it uses a double shot of espresso with micro-frothed milk for a velvety texture.

Caffè Latte: Predominantly steamed milk (2/3) over an espresso base.

Affogato: An Italian dessert consisting of a scoop of vanilla ice cream "drowning" in hot espresso. 4. Study Benefits (Focus Mode) Moderate coffee consumption (between 75mg and 250mg of caffeine) has been shown to improve short-term memory and sustained attention span. In study settings, it aids in the retention of complex concepts and reduces mental fatigue during long sessions. However, to maintain a healthy energy cycle, it is recommended to pair coffee with balanced foods such as avocado toast or yogurt with granola to avoid insulin spikes and maintain stable performance.